PLANO DE UBICACION Y CLAVE DE LAMINAS

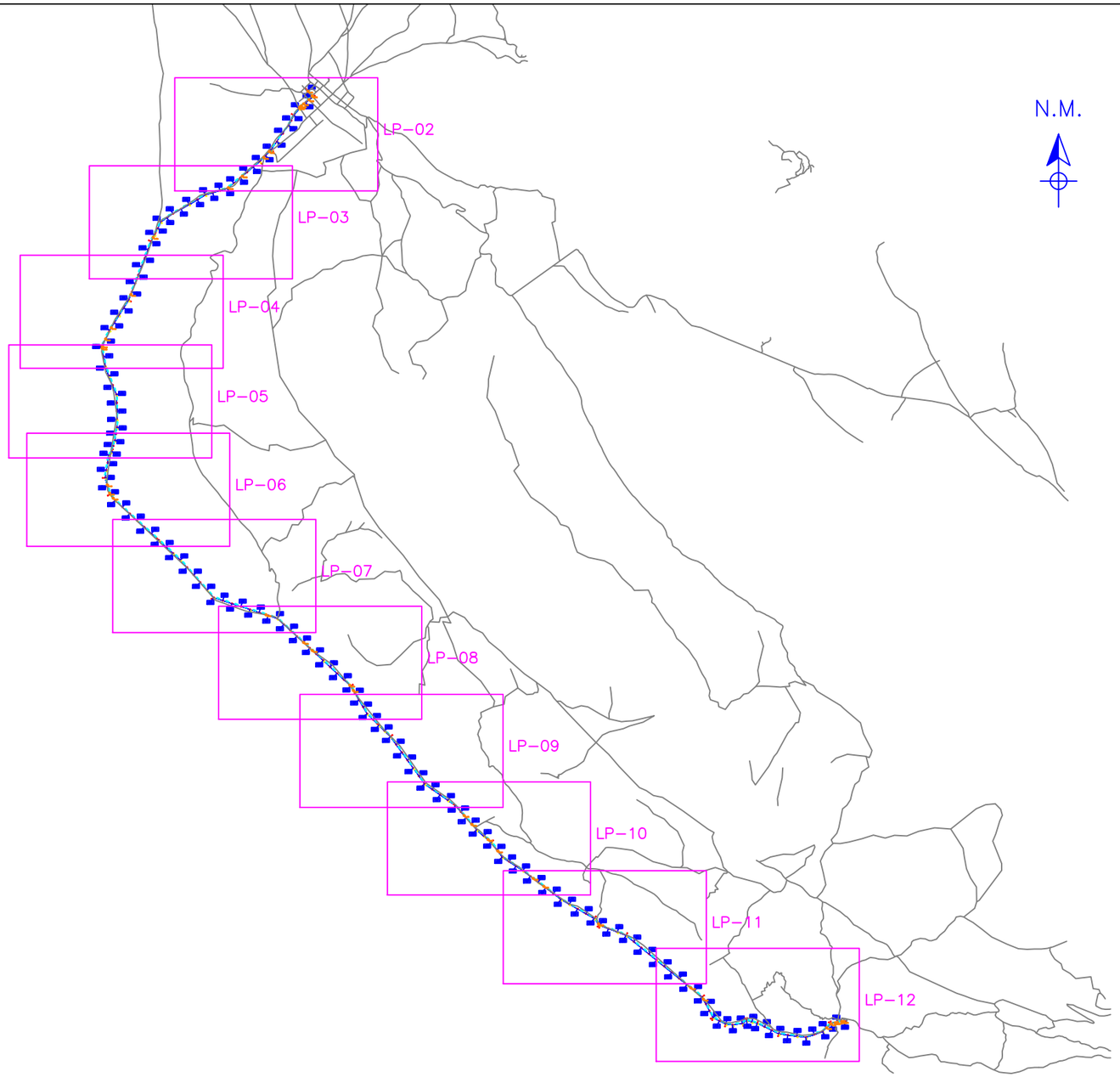
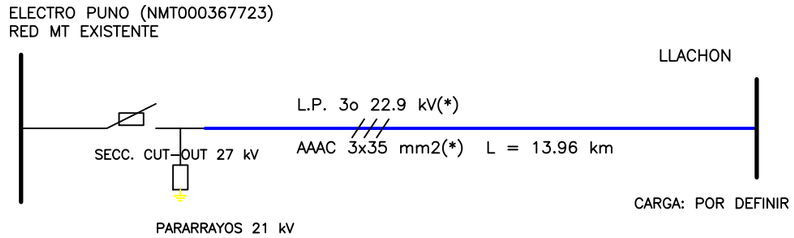


DIAGRAMA UNIFILAR



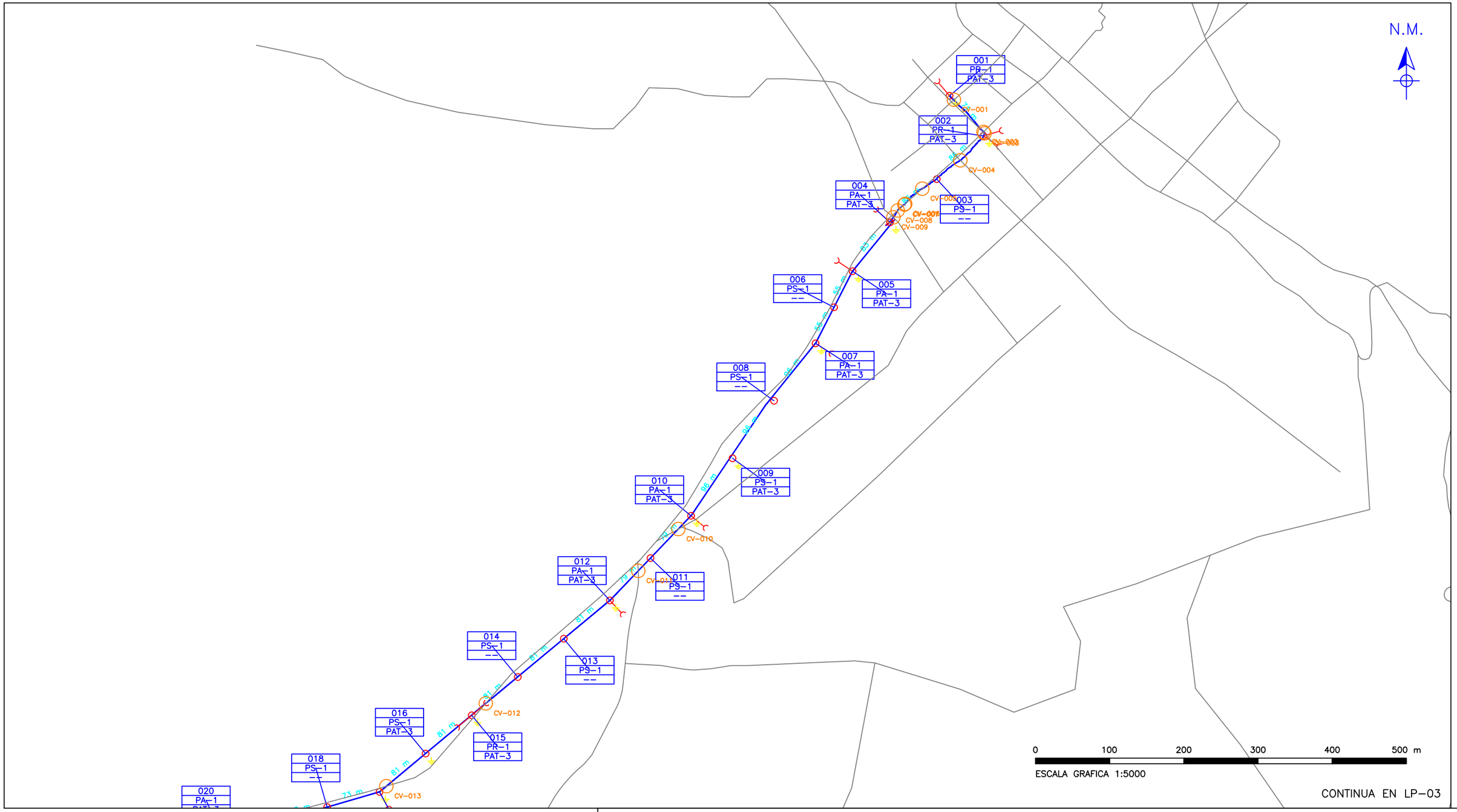
CARACTERISTICAS TECNICAS

Tension nominal: 22.9 kV(*) Sistema: trifasico 3 hilos
Conductor: AAAC 3x35 mm2(*)
Postes: concreto armado centrifugado 12 m
Aislamiento: polimerico (pin y suspension)
Longitud total: 13,956 m Estructuras: 168
Vano max/prom: 97 / 84 m
Normas: CNE Suministro 2011, RD 018–2003–EM/DGE
Coordenadas: UTM WGS84 zona 19S (EPSG:32719)

(*) Tension, conductor y distribucion de postes: hipotesis academica; confirmar con Electro Puno y topografia.

ESCUELA PROFESIONAL DE INGENIERIA MECANICA ELECTRICA – EPIME		UNIVERSIDAD NACIONAL DEL ALTIPLANO – PUNO	
PROYECTO: LINEA PRIMARIA 22.9 kV(*) CAPACHICA – LLACHON			PLANO N
ESCALA 1:50000	FECHA 18/10/2026	TEMA: UBICACION Y DIAGRAMA UNIFILAR	LP–01
UBICACION: REV. 6		DIST. CAPACHICA – PROV. PUNO – REG. PUNO	
REVISADO: _____		AUTOR: POR DESIGNAR	01/13

PLANTA – ESTRUCTURAS E–001 A E–015 | PROG. 0 – 1,121 m



LEYENDA

○	168	POSTE DE C.A.C. 12 m (PROYECTADO)
—(	74	RETENIDA SIMPLE R–1
⏚	84	PUESTA A TIERRA PAT–3
———	13,956 m	CONDUCTOR AAAC 3x35 mm2(*) – 22.9 kV(*)
- - - - -	3	EMPALME ENTRE TRAMOS KML (POR LEVANTAR)
○	44	CRUCE VIAL OSM (VERIFICAR EN CAMPO)

ESCUELA PROFESIONAL DE INGENIERIA MECANICA ELECTRICA – EPIME

UNIVERSIDAD NACIONAL DEL ALTIPLANO – PUNO

PROYECTO: LINEA PRIMARIA 22.9 kV(*) CAPACHICA – LLACHON

PLANO N

LP–02

ESCALA 1:5000

FECHA 18/10/2026

TEMA: RED DE LINEA PRIMARIA – PLANTA

UBICACION: REV. 6

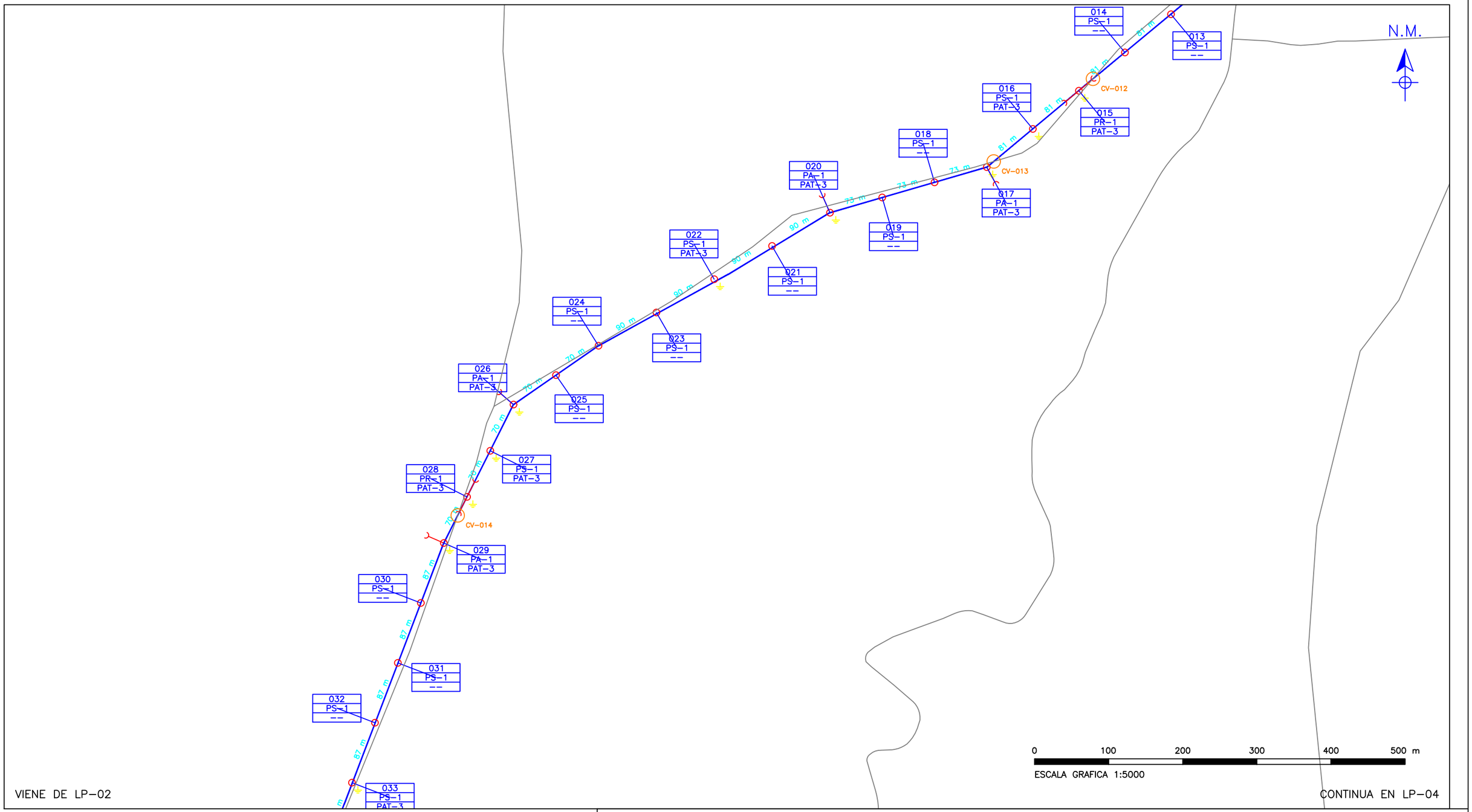
DIST. CAPACHICA – PROV. PUNO – REG. PUNO

REVISADO: _____

AUTOR: POR DESIGNAR

02/13

PLANTA – ESTRUCTURAS E–015 A E–032 | PROG. 1,121 – 2,472 m



LEYENDA

○	168	POSTE DE C.A.C. 12 m (PROYECTADO)
—(	74	RETENIDA SIMPLE R–1
⏚	84	PUESTA A TIERRA PAT–3
———	13,956 m	CONDUCTOR AAAC 3x35 mm2(*) – 22.9 kV(*)
———	3	EMPALME ENTRE TRAMOS KML (POR LEVANTAR)
○	44	CRUCE VIAL OSM (VERIFICAR EN CAMPO)

ESCUELA PROFESIONAL DE INGENIERIA MECANICA ELECTRICA – EPIME

UNIVERSIDAD NACIONAL DEL ALTIPLANO – PUNO

PROYECTO: LINEA PRIMARIA 22.9 kV(*) CAPACHICA – LLACHON

PLANO N

LP–03

ESCALA 1:5000

FECHA 18/10/2026

TEMA: RED DE LINEA PRIMARIA – PLANTA

UBICACION: REV. 6

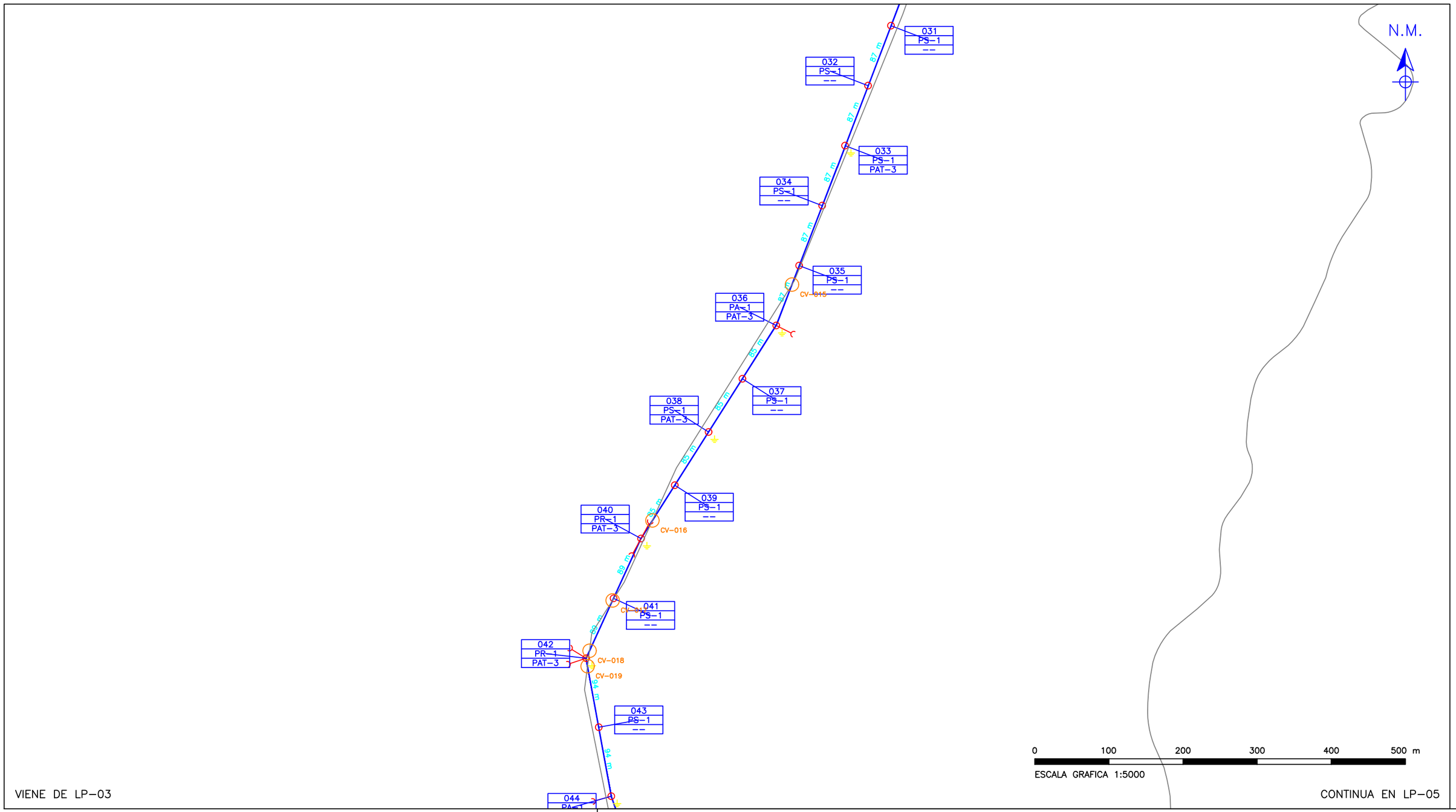
DIST. CAPACHICA – PROV. PUNO – REG. PUNO

REVISADO: _____

AUTOR: POR DESIGNAR

03/13

PLANTA – ESTRUCTURAS E–032 A E–043 | PROG. 2,472 – 3,430 m



LEYENDA

○	168	POSTE DE C.A.C. 12 m (PROYECTADO)
—(	74	RETENIDA SIMPLE R–1
⏚	84	PUESTA A TIERRA PAT–3
———	13,956 m	CONDUCTOR AAAC 3x35 mm2(*) – 22.9 kV(*)
------	3	EMPALME ENTRE TRAMOS KML (POR LEVANTAR)
○	44	CRUCE VIAL OSM (VERIFICAR EN CAMPO)

ESCUELA PROFESIONAL DE INGENIERIA MECANICA ELECTRICA – EPIME — UNIVERSIDAD NACIONAL DEL ALTIPLANO – PUNO

PROYECTO: LINEA PRIMARIA 22.9 kV(*) CAPACHICA – LLACHON

PLANO N: LP–04

ESCALA: 1:5000 | FECHA: 18/10/2026

TEMA: RED DE LINEA PRIMARIA – PLANTA

UBICACION: REV. 6

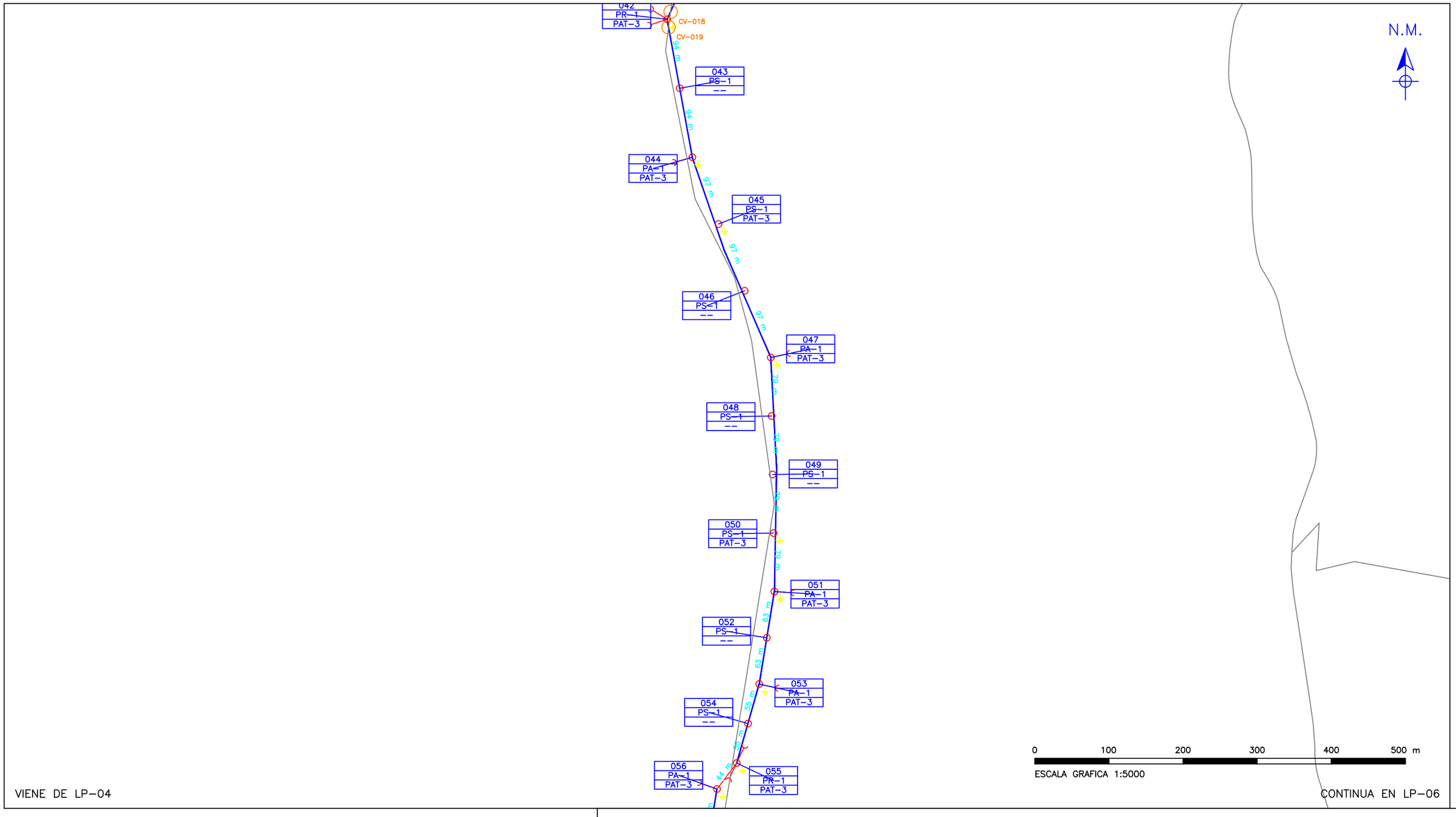
DIST. CAPACHICA – PROV. PUNO – REG. PUNO

REVISADO: _____

AUTOR: POR DESIGNAR

04/13

PLANTA – ESTRUCTURAS E–043 A E–054 | PROG. 3,430 – 4,312 m



LEYENDA

○	168	POSTE DE C.A.C. 12 m (PROYECTADO)
—(	74	RETENIDA SIMPLE R–1
⏚	84	PUESTA A TIERRA PAT–3
———	13,956 m	CONDUCTOR AAAC 3x35 mm2(*) – 22.9 kV(*)
------	3	EMPALME ENTRE TRAMOS KML (POR LEVANTAR)
○	44	CRUCE VIAL OSM (VERIFICAR EN CAMPO)

ESCUELA PROFESIONAL DE INGENIERIA MECANICA ELECTRICA – EPIME — UNIVERSIDAD NACIONAL DEL ALTIPLANO – PUNO

PROYECTO: LINEA PRIMARIA 22.9 kV(*) CAPACHICA – LLACHON

PLANO N: LP–05

ESCALA: 1:5000

FECHA: 18/10/2026

TEMA: RED DE LINEA PRIMARIA – PLANTA

UBICACION: REV. 6

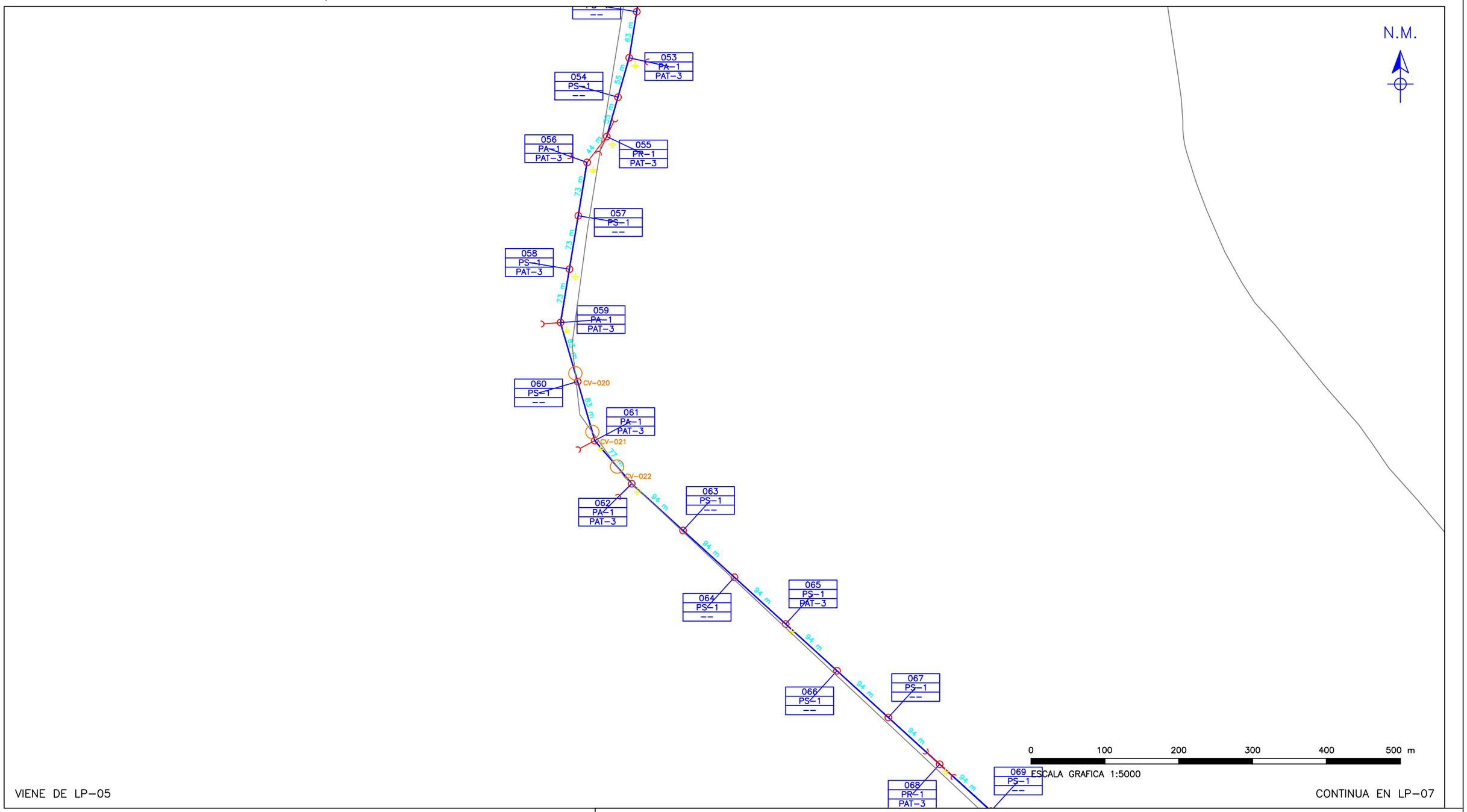
DIST. CAPACHICA – PROV. PUNO – REG. PUNO

REVISADO: _____

AUTOR: POR DESIGNAR

05/13

PLANTA – ESTRUCTURAS E–054 A E–067 | PROG. 4,312 – 5,344 m



VIENE DE LP–05

CONTINUA EN LP–07

LEYENDA

○	168	POSTE DE C.A.C. 12 m (PROYECTADO)
—(—	74	RETENIDA SIMPLE R–1
⏚	84	PUESTA A TIERRA PAT–3
———	13,956 m	CONDUCTOR AAAC 3x35 mm2(*) – 22.9 kV(*)
– – – –	3	EMPALME ENTRE TRAMOS KML (POR LEVANTAR)
○	44	CRUCE VIAL OSM (VERIFICAR EN CAMPO)

ESCUELA PROFESIONAL DE INGENIERIA MECANICA ELECTRICA – EPIME

UNIVERSIDAD NACIONAL DEL ALTIPLANO – PUNO

PROYECTO: LINEA PRIMARIA 22.9 kV(*) CAPACHICA – LLACHON

PLANO N

LP–06

ESCALA 1:5000

FECHA 18/10/2026

TEMA: RED DE LINEA PRIMARIA – PLANTA

UBICACION: REV. 6

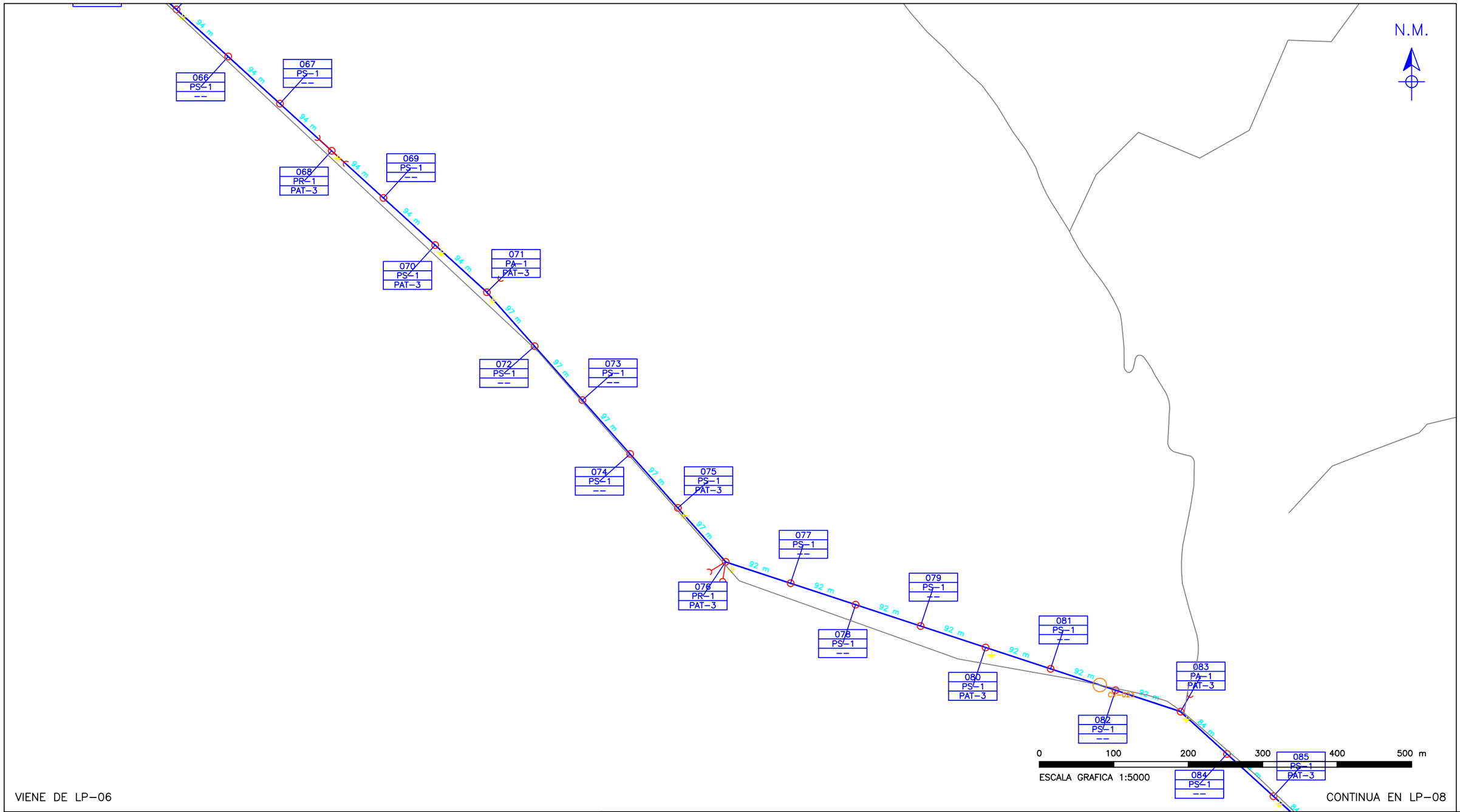
DIST. CAPACHICA – PROV. PUNO – REG. PUNO

REVISADO: _____

AUTOR: POR DESIGNAR

06/13

PLANTA – ESTRUCTURAS E–067 A E–083 | PROG. 5,344 – 6,846 m



VIENE DE LP–06

CONTINUA EN LP–08

LEYENDA

○	168	POSTE DE C.A.C. 12 m (PROYECTADO)
—(	74	RETENIDA SIMPLE R–1
⏚	84	PUESTA A TIERRA PAT–3
———	13,956 m	CONDUCTOR AAAC 3x35 mm2(*) – 22.9 kV(*)
– – – –	3	EMPALME ENTRE TRAMOS KML (POR LEVANTAR)
○	44	CRUCE VIAL OSM (VERIFICAR EN CAMPO)

ESCUELA PROFESIONAL DE INGENIERIA MECANICA ELECTRICA – EPIME

UNIVERSIDAD NACIONAL DEL ALTIPLANO – PUNO

PROYECTO: LINEA PRIMARIA 22.9 kV(*) CAPACHICA – LLACHON

PLANO N

LP–07

ESCALA 1:5000

FECHA 18/10/2026

TEMA: RED DE LINEA PRIMARIA – PLANTA

UBICACION: REV. 6

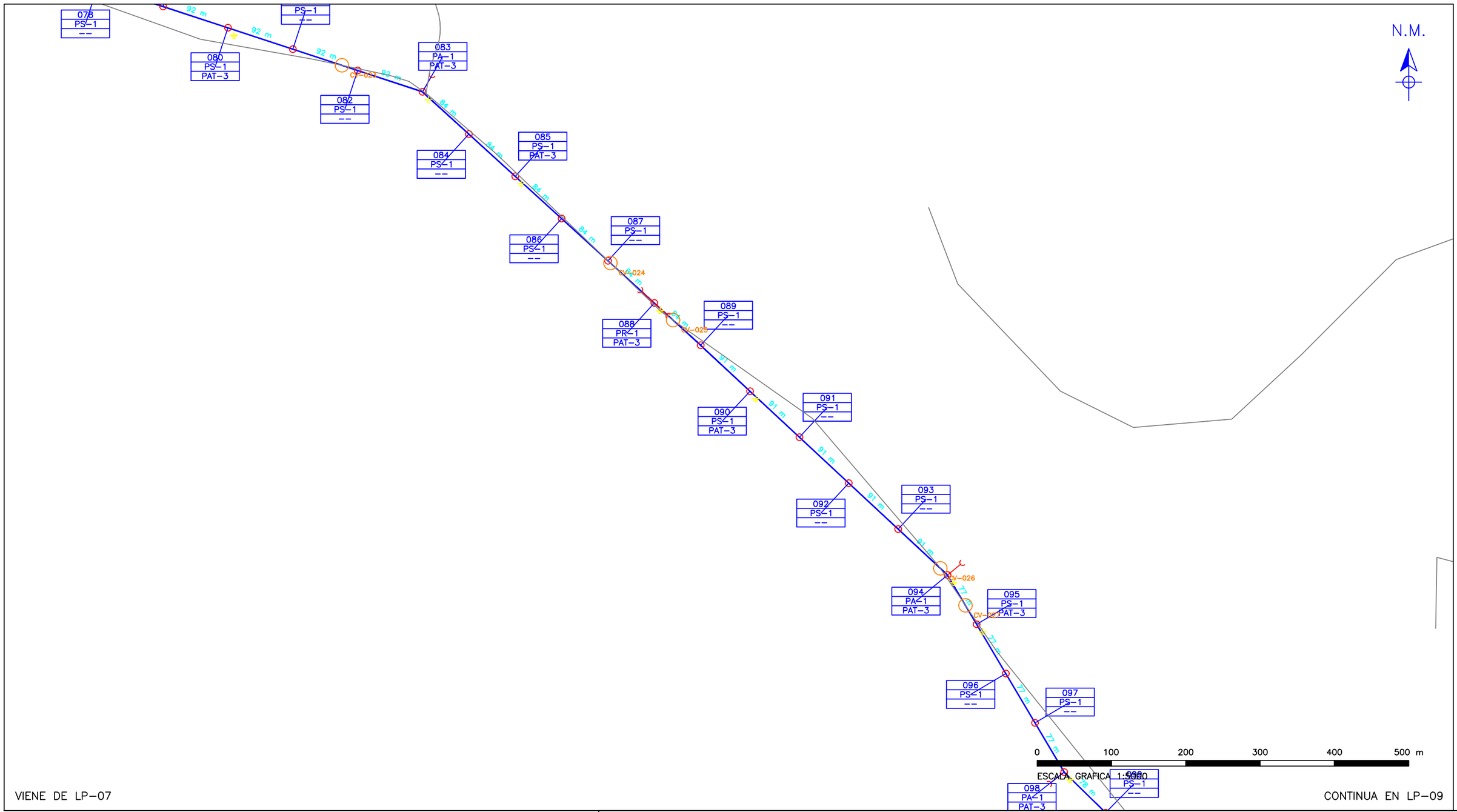
DIST. CAPACHICA – PROV. PUNO – REG. PUNO

REVISADO: _____

AUTOR: POR DESIGNAR

07/13

PLANTA – ESTRUCTURAS E–083 A E–097 | PROG. 6,846 – 8,037 m



LEYENDA

○	168	POSTE DE C.A.C. 12 m (PROYECTADO)
—(	74	RETENIDA SIMPLE R–1
⏚	84	PUESTA A TIERRA PAT–3
———	13,956 m	CONDUCTOR AAAC 3x35 mm2(*) – 22.9 kV(*)
– – –	3	EMPALME ENTRE TRAMOS KML (POR LEVANTAR)
○	44	CRUCE VIAL OSM (VERIFICAR EN CAMPO)

ESCUELA PROFESIONAL DE INGENIERIA MECANICA ELECTRICA – EPIME

UNIVERSIDAD NACIONAL DEL ALTIPLANO – PUNO

PROYECTO: LINEA PRIMARIA 22.9 kV(*) CAPACHICA – LLACHON

PLANO N

ESCALA 1:5000

FECHA 18/10/2026

TEMA: RED DE LINEA PRIMARIA – PLANTA

LP–08

UBICACION: REV. 6

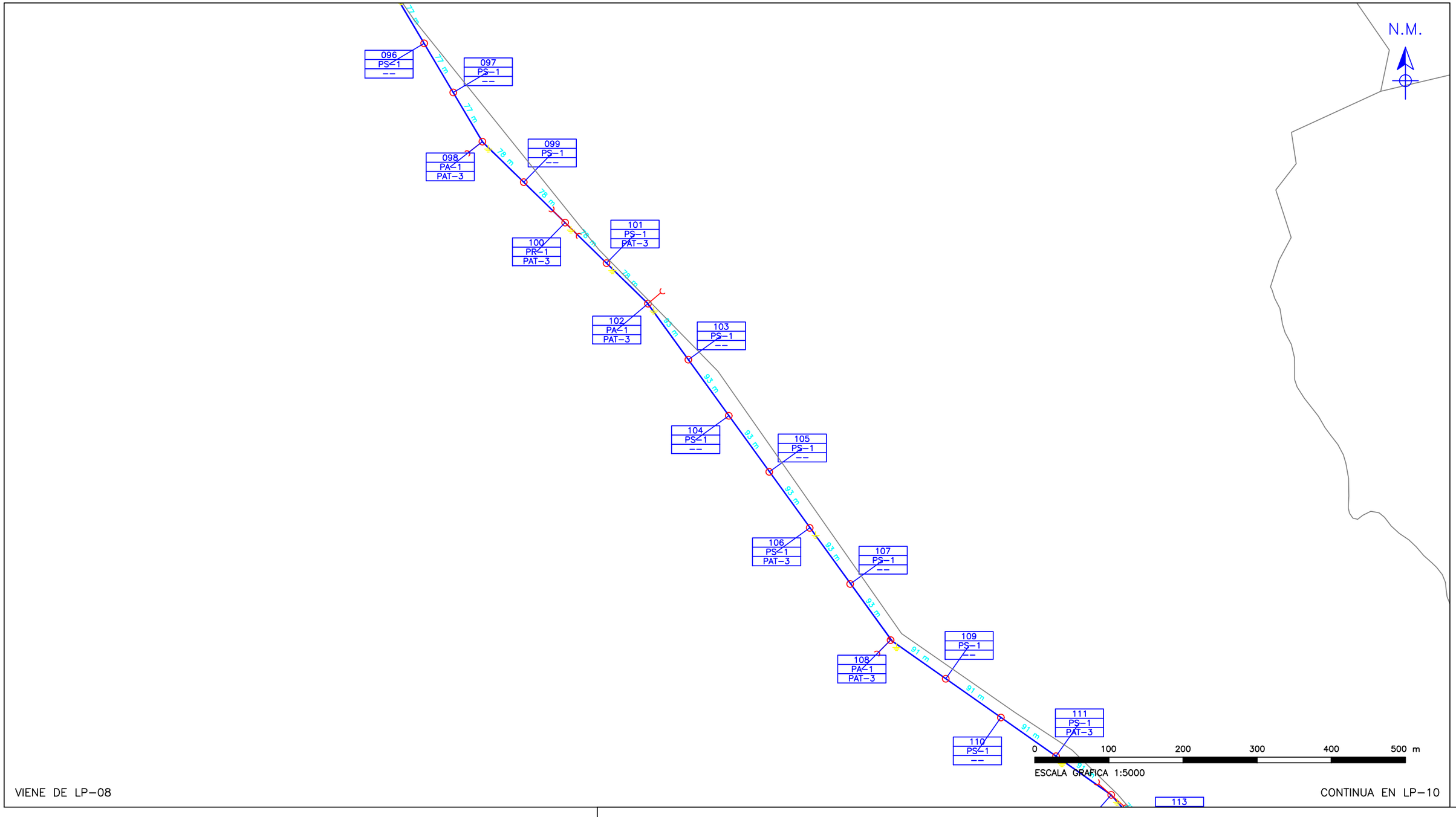
DIST. CAPACHICA – PROV. PUNO – REG. PUNO

REVISADO: _____

AUTOR: POR DESIGNAR

08/13

PLANTA – ESTRUCTURAS E–097 A E–110 | PROG. 8,037 – 9,168 m



LEYENDA

○	168	POSTE DE C.A.C. 12 m (PROYECTADO)
—(	74	RETENIDA SIMPLE R–1
⏚	84	PUESTA A TIERRA PAT–3
———	13,956 m	CONDUCTOR AAAC 3x35 mm2(*) – 22.9 kV(*)
– – –	3	EMPALME ENTRE TRAMOS KML (POR LEVANTAR)
○	44	CRUCE VIAL OSM (VERIFICAR EN CAMPO)

ESCUELA PROFESIONAL DE INGENIERIA MECANICA ELECTRICA – EPIME — UNIVERSIDAD NACIONAL DEL ALTIPLANO – PUNO

PROYECTO: LINEA PRIMARIA 22.9 kV(*) CAPACHICA – LLACHON

ESCALA 1:5000 | FECHA 18/10/2026

TEMA: RED DE LINEA PRIMARIA – PLANTA

UBICACION: REV. 6

DIST. CAPACHICA – PROV. PUNO – REG. PUNO

REVISADO: _____

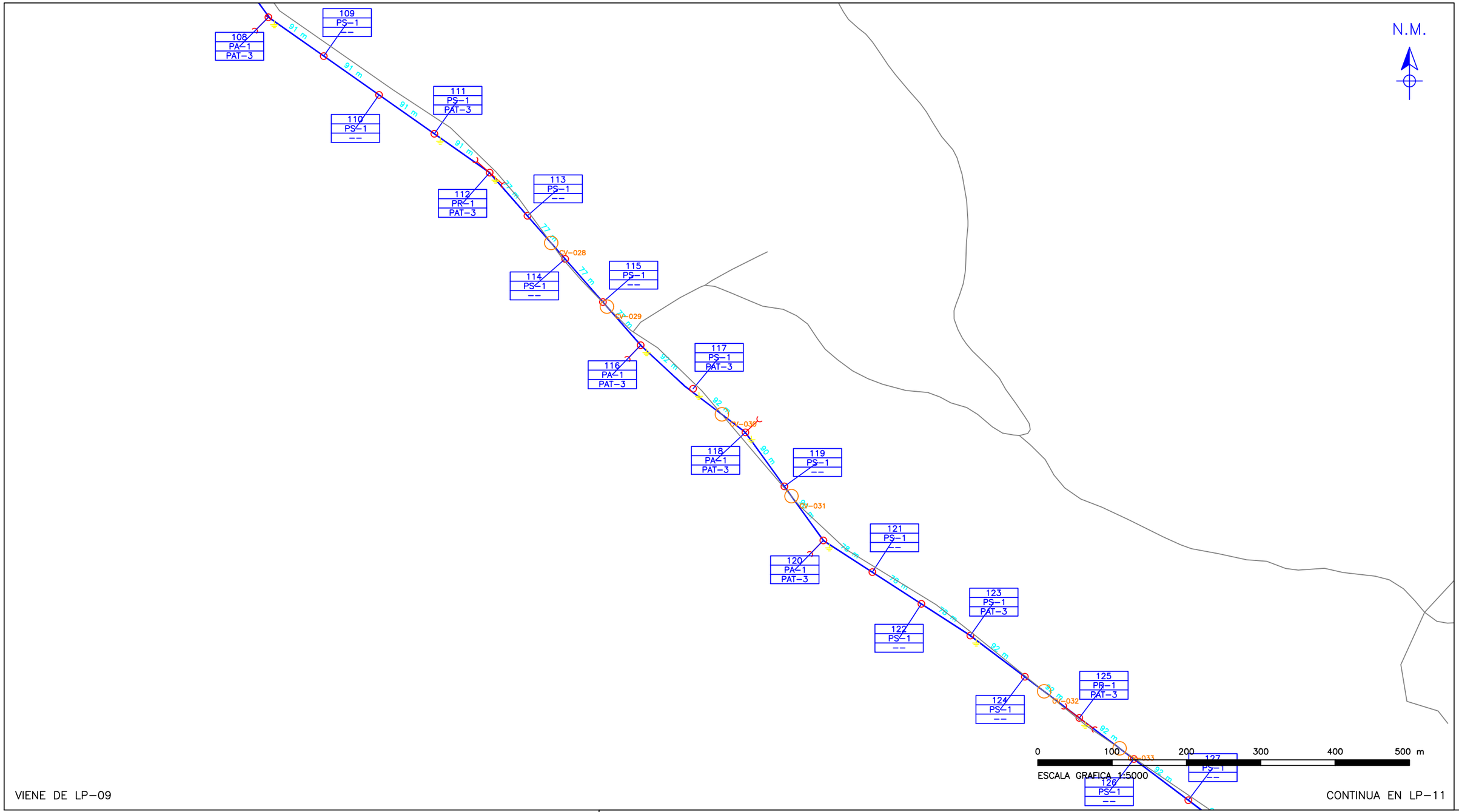
AUTOR: POR DESIGNAR

PLANO N

LP–09

09/13

PLANTA – ESTRUCTURAS E–110 A E–125 | PROG. 9,168 – 10,440 m



LEYENDA

○	168	POSTE DE C.A.C. 12 m (PROYECTADO)
—(—	74	RETENIDA SIMPLE R–1
⏚	84	PUESTA A TIERRA PAT–3
———	13,956 m	CONDUCTOR AAAC 3x35 mm2(*) – 22.9 kV(*)
– – –	3	EMPALME ENTRE TRAMOS KML (POR LEVANTAR)
○	44	CRUCE VIAL OSM (VERIFICAR EN CAMPO)

ESCUELA PROFESIONAL DE INGENIERIA MECANICA ELECTRICA – EPIME — UNIVERSIDAD NACIONAL DEL ALTIPLANO – PUNO

PROYECTO: LINEA PRIMARIA 22.9 kV(*) CAPACHICA – LLACHON

PLANO N: LP–10

ESCALA: 1:5000 | FECHA: 18/10/2026

TEMA: RED DE LINEA PRIMARIA – PLANTA

UBICACION: REV. 6

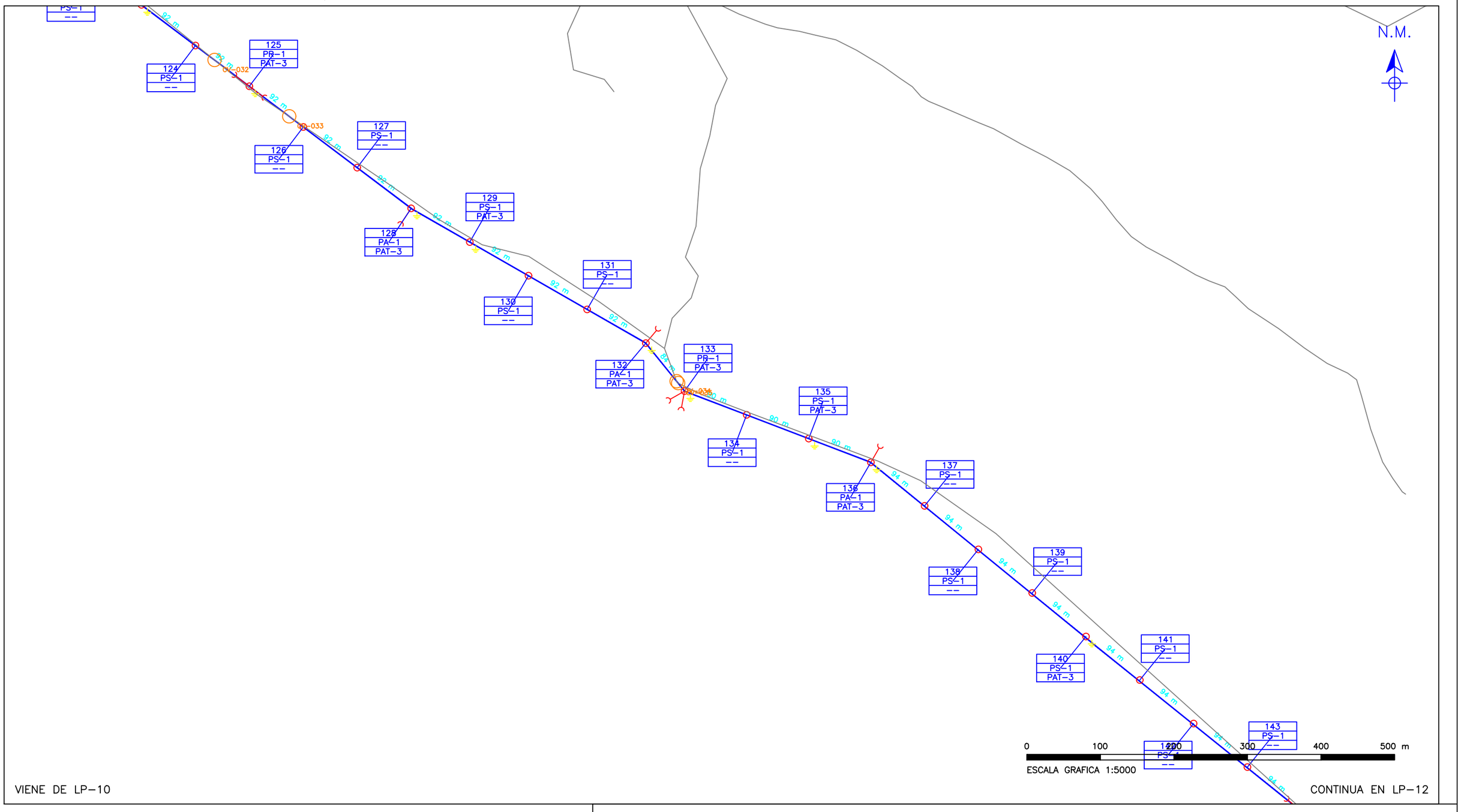
DIST. CAPACHICA – PROV. PUNO – REG. PUNO

REVISADO: _____

AUTOR: POR DESIGNAR

10/13

PLANTA – ESTRUCTURAS E–125 A E–142 | PROG. 10,440 – 12,002 m



VIENE DE LP–10

CONTINUA EN LP–12

LEYENDA

○	168	POSTE DE C.A.C. 12 m (PROYECTADO)
—(—	74	RETENIDA SIMPLE R–1
⏚	84	PUESTA A TIERRA PAT–3
———	13,956 m	CONDUCTOR AAAC 3x35 mm2(*) – 22.9 kV(*)
------	3	EMPALME ENTRE TRAMOS KML (POR LEVANTAR)
○	44	CRUCE VIAL OSM (VERIFICAR EN CAMPO)

ESCUELA PROFESIONAL DE INGENIERIA MECANICA ELECTRICA – EPIME

UNIVERSIDAD NACIONAL DEL ALTIPLANO – PUNO

PROYECTO: LINEA PRIMARIA 22.9 kV(*) CAPACHICA – LLACHON

PLANO N

LP–11

ESCALA 1:5000

FECHA 18/10/2026

TEMA: RED DE LINEA PRIMARIA – PLANTA

UBICACION: REV. 6

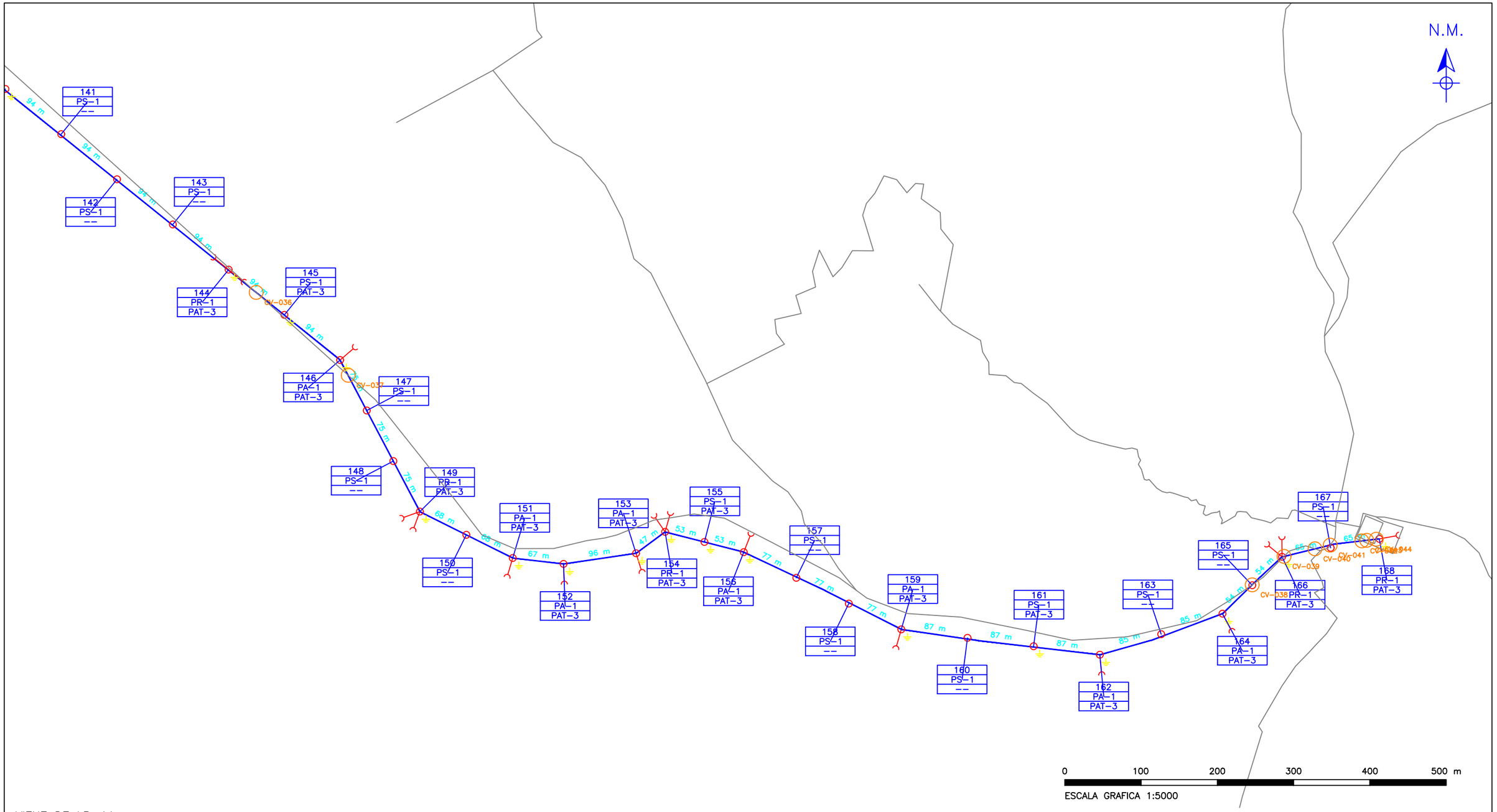
DIST. CAPACHICA – PROV. PUNO – REG. PUNO

REVISADO: _____

AUTOR: POR DESIGNAR

11/13

PLANTA – ESTRUCTURAS E–142 A E–168 | PROG. 12,002 – 13,955 m



VIENE DE LP–11

LEYENDA

○	168	POSTE DE C.A.C. 12 m (PROYECTADO)
—(	74	RETENIDA SIMPLE R–1
⏚	84	PUESTA A TIERRA PAT–3
———	13,956 m	CONDUCTOR AAAC 3x35 mm2(*) – 22.9 kV(*)
– – –	3	EMPALME ENTRE TRAMOS KML (POR LEVANTAR)
○	44	CRUCE VIAL OSM (VERIFICAR EN CAMPO)

ESCUELA PROFESIONAL DE INGENIERIA MECANICA ELECTRICA – EPIME

UNIVERSIDAD NACIONAL DEL ALTIPLANO – PUNO

PROYECTO: LINEA PRIMARIA 22.9 kV(*) CAPACHICA – LLACHON

PLANO N

LP–12

ESCALA 1:5000

FECHA 18/10/2026

TEMA: RED DE LINEA PRIMARIA – PLANTA

UBICACION: REV. 6

DIST. CAPACHICA – PROV. PUNO – REG. PUNO

REVISADO: _____

AUTOR: POR DESIGNAR

12/13

CUADRO DE ESTRUCTURAS – COORDENADAS UTM WGS84 19S (PROPUESTA ACADEMICA, POR REPLANTEAR)

N	ARM.	POSTE	RET.	PAT	PROG.	VANO	ESTE	NORTE
001	PR–1	12/500	1xR–1	PAT–3	0	0.0	410865.19	8270474.21
002	PR–1	12/500	2xR–1	PAT–3	71	70.9	410910.89	8270420.04
003	PS–1	12/300	–	–	157	85.7	410847.97	8270361.80
004	PA–1	12/400	1xR–1	PAT–3	242	85.7	410785.05	8270303.57
005	PA–1	12/400	1xR–1	PAT–3	325	83.0	410734.43	8270237.83
006	PS–1	12/300	–	–	380	54.6	410709.52	8270189.25
007	PA–1	12/400	1xR–1	PAT–3	434	54.6	410684.61	8270140.67
008	PS–1	12/300	–	–	530	95.5	410628.66	8270063.23
009	PS–1	12/300	–	PAT–3	626	95.5	410572.71	8269985.79
010	PA–1	12/400	1xR–1	PAT–3	721	95.5	410516.76	8269908.35
011	PS–1	12/300	–	–	800	79.0	410462.11	8269851.30
012	PA–1	12/400	1xR–1	PAT–3	879	79.0	410407.46	8269794.25
013	PS–1	12/300	–	–	960	80.7	410345.35	8269742.67
014	PS–1	12/300	–	–	1041	80.7	410283.24	8269691.09
015	PR–1	12/500	2xR–1	PAT–3	1121	80.7	410221.13	8269639.51
016	PS–1	12/300	–	PAT–3	1202	80.7	410159.03	8269587.92
017	PA–1	12/400	1xR–1	PAT–3	1283	80.7	410096.92	8269536.34
018	PS–1	12/300	–	–	1356	73.4	410026.38	8269515.88
019	PS–1	12/300	–	–	1430	73.4	409955.84	8269495.41
020	PA–1	12/400	1xR–1	PAT–3	1503	73.4	409885.31	8269474.95
021	PS–1	12/300	–	–	1593	90.0	409807.27	8269430.06
022	PS–1	12/300	–	PAT–3	1683	90.0	409729.23	8269385.17
023	PS–1	12/300	–	–	1773	90.0	409651.19	8269340.28
024	PS–1	12/300	–	–	1863	90.0	409573.15	8269295.39
025	PS–1	12/300	–	–	1933	69.8	409515.76	8269255.69
026	PA–1	12/400	1xR–1	PAT–3	2003	69.8	409458.37	8269216.00
027	PS–1	12/300	–	PAT–3	2072	69.7	409427.05	8269153.75
028	PR–1	12/500	2xR–1	PAT–3	2142	69.7	409395.72	8269091.49
029	PA–1	12/400	1xR–1	PAT–3	2212	69.7	409364.39	8269029.23
030	PS–1	12/300	–	–	2298	86.6	409333.43	8268948.40
031	PS–1	12/300	–	–	2385	86.6	409302.46	8268867.57
032	PS–1	12/300	–	–	2472	86.6	409271.50	8268786.74
033	PS–1	12/300	–	PAT–3	2558	86.6	409240.53	8268705.91
034	PS–1	12/300	–	–	2645	86.6	409209.57	8268625.08
035	PS–1	12/300	–	–	2731	86.6	409178.60	8268544.24
036	PA–1	12/400	1xR–1	PAT–3	2818	86.6	409147.64	8268463.41
037	PS–1	12/300	–	–	2903	85.0	409102.06	8268391.62
038	PS–1	12/300	–	PAT–3	2988	85.0	409056.48	8268319.83
039	PS–1	12/300	–	–	3073	85.0	409010.89	8268248.04
040	PR–1	12/500	2xR–1	PAT–3	3158	85.0	408965.31	8268176.25
041	PS–1	12/300	–	–	3247	88.8	408928.29	8268095.53
042	PR–1	12/500	2xR–1	PAT–3	3336	88.8	408891.27	8268014.80
043	PS–1	12/300	–	–	3430	94.5	408908.25	8267921.80
044	PA–1	12/400	1xR–1	PAT–3	3525	94.5	408925.22	8267828.79
045	PS–1	12/300	–	PAT–3	3621	96.6	408960.51	8267738.82
046	PS–1	12/300	–	–	3718	96.6	408995.80	8267648.85
047	PA–1	12/400	1xR–1	PAT–3	3815	96.6	409031.09	8267558.88
048	PS–1	12/300	–	–	3893	78.9	409032.30	8267480.02
049	PS–1	12/300	–	–	3972	78.9	409033.51	8267401.15
050	PS–1	12/300	–	PAT–3	4051	78.9	409034.72	8267322.28
051	PA–1	12/400	1xR–1	PAT–3	4130	78.9	409035.93	8267243.41
052	PS–1	12/300	–	–	4193	63.2	409025.69	8267181.07
053	PA–1	12/400	1xR–1	PAT–3	4256	63.2	409015.45	8267118.72
054	PS–1	12/300	–	–	4312	55.4	409000.25	8267065.47
055	PR–1	12/500	2xR–1	PAT–3	4367	55.4	408985.05	8267012.21
056	PA–1	12/400	1xR–1	PAT–3	4411	44.0	408958.49	8266977.18
057	PS–1	12/300	–	–	4484	73.2	408946.51	8266904.95
058	PS–1	12/300	–	PAT–3	4558	73.2	408934.52	8266832.73
059	PA–1	12/400	1xR–1	PAT–3	4631	73.2	408922.54	8266760.50
060	PS–1	12/300	–	–	4714	83.2	408945.72	8266680.63
061	PA–1	12/400	1xR–1	PAT–3	4797	83.2	408968.90	8266600.76
062	PA–1	12/400	1xR–1	PAT–3	4874	76.9	409018.93	8266542.39
063	PS–1	12/300	–	–	4968	94.0	409088.38	8266479.11
064	PS–1	12/300	–	–	5062	94.0	409157.84	8266415.83
065	PS–1	12/300	–	PAT–3	5156	94.0	409227.29	8266352.56
066	PS–1	12/300	–	–	5250	94.0	409296.74	8266289.28
067	PS–1	12/300	–	–	5344	94.0	409366.20	8266226.01
068	PR–1	12/500	2xR–1	PAT–3	5438	94.0	409435.65	8266162.73
069	PS–1	12/300	–	–	5532	94.0	409505.11	8266099.46
070	PS–1	12/300	–	PAT–3	5626	94.0	409574.56	8266036.18
071	PA–1	12/400	1xR–1	PAT–3	5720	94.0	409644.02	8265972.90
072	PS–1	12/300	–	–	5816	96.7	409708.14	8265900.53
073	PS–1	12/300	–	–	5913	96.7	409772.26	8265828.16
074	PS–1	12/300	–	–	6010	96.7	409836.38	8265755.79
075	PS–1	12/300	–	PAT–3	6106	96.7	409900.50	8265683.42
076	PR–1	12/500	2xR–1	PAT–3	6203	96.7	409964.62	8265611.04
077	PS–1	12/300	–	–	6295	91.9	410051.89	8265582.32
078	PS–1	12/300	–	–	6387	91.9	410139.16	8265553.60
079	PS–1	12/300	–	–	6479	91.9	410226.43	8265524.88
080	PS–1	12/300	–	PAT–3	6571	91.9	410313.70	8265496.16
081	PS–1	12/300	–	–	6662	91.9	410400.97	8265467.44
082	PS–1	12/300	–	–	6754	91.9	410488.25	8265438.72
083	PA–1	12/400	1xR–1	PAT–3	6846	91.9	410575.52	8265410.00
084	PS–1	12/300	–	–	6931	84.4	410637.89	8265353.19
085	PS–1	12/300	–	PAT–3	7015	84.4	410700.26	8265296.37
086	PS–1	12/300	–	–	7099	84.4	410762.63	8265239.56
087	PS–1	12/300	–	–	7184	84.4	410825.01	8265182.74
088	PR–1	12/500	2xR–1	PAT–3	7268	84.4	410887.38	8265125.93
089	PS–1	12/300	–	–	7352	84.4	410949.75	8265069.12
090	PS–1	12/300	–	PAT–3	7443	90.7	411016.18	8265007.29
091	PS–1	12/300	–	–	7534	90.7	411082.61	8264945.47
092	PS–1	12/300	–	–	7625	90.7	411149.04	8264883.65
093	PS–1	12/300	–	–	7715	90.7	411215.46	8264821.83
094	PA–1	12/400	1xR–1	PAT–3	7806	90.7	411281.89	8264760.00
095	PS–1	12/300	–	PAT–3	7883	77.0	411321.12	8264693.73
096	PS–1	12/300	–	–	7960	77.0	411360.35	8264627.46
097	PS–1	12/300	–	–	8037	77.0	411399.58	8264561.19
098	PA–1	12/400	1xR–1	PAT–3	8114	77.0	411438.82	8264494.92
099	PS–1	12/300	–	–	8192	78.1	411494.64	8264440.29
100	PR–1	12/500	2xR–1	PAT–3	8270	78.1	411550.46	8264385.66
101	PS–1	12/300	–	PAT–3	8349	78.1	411606.28	8264331.02
102	PA–1	12/400	1xR–1	PAT–3	8427	78.1	411662.10	8264276.39
103	PS–1	12/300	–	–	8520	93.3	411716.65	8264200.75
104	PS–1	12/300	–	–	8613	93.3	411771.20	8264125.11
105	PS–1	12/300	–	–	8706	93.3	411825.75	8264049.47
106	PS–1	12/300	–	PAT–3	8800	93.3	411880.30	8263973.83
107	PS–1	12/300	–	–	8893	93.3	411934.85	8263898.19
108	PA–1	12/400	1xR–1	PAT–3	8986	93.3	411989.41	8263822.55
109	PS–1	12/300	–	–	9077	90.9	412063.79	8263770.34
110	PS–1	12/300	–	–	9168	90.9	412138.18	8263718.12
111	PS–1	12/300	–	PAT–3	9259	90.9	412212.57	8263665.90
112	PR–1	12/500	2xR–1	PAT–3	9350	90.9	412286.96	8263613.69
113	PS–1	12/300	–	–	9427	77.1	412337.74	8263555.69
114	PS–1	12/300	–	–	9504	77.1	412388.53	8263497.69
115	PS–1	12/300	–	–	9581	77.1	412439.31	8263439.70
116	PA–1	12/400	1xR–1	PAT–3	9658	77.1	412490.09	8263381.70
117	PS–1	12/300	–	PAT–3	9750	91.6	412560.51	8263323.14
118	PA–1	12/400	1xR–1	PAT–3	9841	91.6	412630.94	8263264.58
119	PS–1	12/300	–	–	9931	89.6	412683.33	8263191.92
120	PA–1	12/400	1xR–1	PAT–3	10020	89.6	412735.73	8263119.27
121	PS–1	12/300	–	–	10099	78.5	412801.63	8263076.65
122	PS–1	12/300	–	–	10177	78.5	412867.52	8263034.02
123	PS–1	12/300	–	PAT–3	10256	78.5	412933.42	8262991.40
124	PS–1	12/300	–	–	10348	91.8	413006.70	8262936.06
125	PR–1	12/500	2xR–1	PAT–3	10440	91.8	413079.98	8262880.73
126	PS–1	12/300	–	–	10531	91.8	413153.26	8262825.40
127	PS–1	12/300	–	–	10623	91.8	413226.54	8262770.06
128	PA–1	12/400	1xR–1	PAT–3	10715	91.8	413299.82	8262714.73
129	PS–1	12/300	–	PAT–3	10807	92.0	413379.58	8262668.95
130	PS–1	12/300	–	–	10899	92.0	413459.33	8262623.17
131	PS–1	12/300	–	–	10991	92.0	413539.09	8262577.39
132	PA–1	12/400	1xR–1	PAT–3	11083	92.0	413618.84	8262531.61
133	PR–1	12/500	2xR–1	PAT–3	11167	83.8	413671.48	8262466.37
134	PS–1	12/300	–	–	11257	90.3	413755.86	8262434.09
135	PS–1	12/300	–	PAT–3	11347	90.3	413840.24	8262401.80
136	PA–1	12/400	1xR–1	PAT–3	11438	90.3	413924.63	8262369.52
137	PS–1	12/300	–	–	11532	94.1	413997.74	8262310.34
138	PS–1	12/300	–	–	11626	94.1	414070.86	8262251.16
139	PS–1	12/300	–	–	11720	94.1	414143.97	8262191.98
140	PS–1	12/300	–	PAT–3	11814	94.1	414217.09	8262132.80
141	PS–1	12/300	–	–	11908	94.1	414290.21	8262073.61
142	PS–1	12/300	–	–	12002	94.1	414363.32	8262014.43
143	PS–1	12/300	–	–	12096	94.1	414436.44	8261955.25
144	PR–1	12/500	2xR–1	PAT–3	12190	94.1	414509.56	8261896.07
145	PS–1	12/300	–	PAT–3	12284	94.1	414582.67	8261836.89
146	PA–1	12/400	1xR–1	PAT–3	12378	94.1	414655.79	8261777.71
147	PS–1	12/300	–	–	12453	74.8	414690.62	8261711.51
148	PS–1	12/300	–	–	12528	74.8	414725.46	8261645.31
149	PR–1	12/500	2xR–1	PAT–3	12603	74.8	414760.30	8261579.11
150	PS–1	12/300	–	–	12671	68.1	414821.17	8261548.68
151	PA–1	12/400	1xR–1	PAT–3	12739	68.1	414882.04	8261518.25
152	PA–1	12/400	1xR–1	PAT–3	12806	67.0	414948.60	8261510.53
153	PA–1	12/400	1xR–1	PAT–3	12902	96.1	415043.63	8261524.50
154	PR–1	12/500	2xR–1	PAT–3	12949	47.3	415081.74	8261552.57
155	PS–1	12/300	–	PAT–3	13002	53.2	415133.27	8261539.32
156	PA–1	12/400	1xR–1	PAT–3	13056	53.2	415184.80	8261526.07
157	PS–1	12/300	–	–	13132	76.7	415253.63	8261492.33
158	PS–1	12/300	–	–	13209	76.7	415322.46	8261458.59
159	PA–1	12/400	1xR–1	PAT–3	13286	76.7	415391.29	8261424.85
160	PS–1	12/300	–	–	13373	87.4	415478.02	8261413.78
161	PS–1	12/300	–	PAT–3	13460	87.4	415564.75	8261402.71
162	PA–1	12/400	1xR–1	PAT–3	13548	87.4	415651.49	8261391.63
163	PS–1	12/300	–	–	13633	84.8	415731.88	8261418.49
164	PA–1	12/400	1xR–1	PAT–3	13717	84.8	415812.28	8261445.35
165	PS–1	12/300	–	–	13771	54.0	415851.31	8261482.59
166	PR–1	12/500	2xR–1	PAT–3	13825	54.0	415890.35	8261519.84
167	PS–1	12/300	–	–	13890	64.8	415954.06	8261531.60
168	PR–1	12/500	1xR–1	PAT–3	13955	64.8	416017.77	8261543.36

CUADRO DE ESTRUCTURAS

TIPO	ARMADO	POSTE	RETENIDA	PAT	CANT.	DESCRIPCION
Angulo	PA–1	12/400	1xR–1	PAT–3	38	Angulo 5–30, cadena suspension
Anclaje de paso	PR–1	12/500	2xR–1	PAT–3	10	Retencion, doble cadena anclaje
Angulo	PR–1	12/500	2xR–1	PAT–3	7	Retencion, doble cadena anclaje
Terminal	PR–1	12/500	1xR–1	PAT–3	2	Retencion, doble cadena anclaje
Alineamiento	PS–1	12/300	–	–	84	Alineamiento 0–5, aislador pin
Alineamiento	PS–1	12/300	–	PAT–3	27	Alineamiento 0–5, aislador pin
			TOTAL DE POSTES		168	

ESCUELA PROFESIONAL DE INGENIERIA MECANICA ELECTRICA – EPIME

UNIVERSIDAD NACIONAL DEL ALTIPLANO – PUNO

PROYECTO: LINEA PRIMARIA 22.9 kV(*) CAPACHICA – LLACHON

PLANO N

LP–13

ESCALA S/E

FECHA 18/10/2026

TEMA: CUADRO DE ESTRUCTURAS Y COORDENADAS

UBICACION: REV. 6

DIST. CAPACHICA – PROV. PUNO – REG. PUNO

REVISADO: _____

AUTOR: POR DESIGNAR

13/13